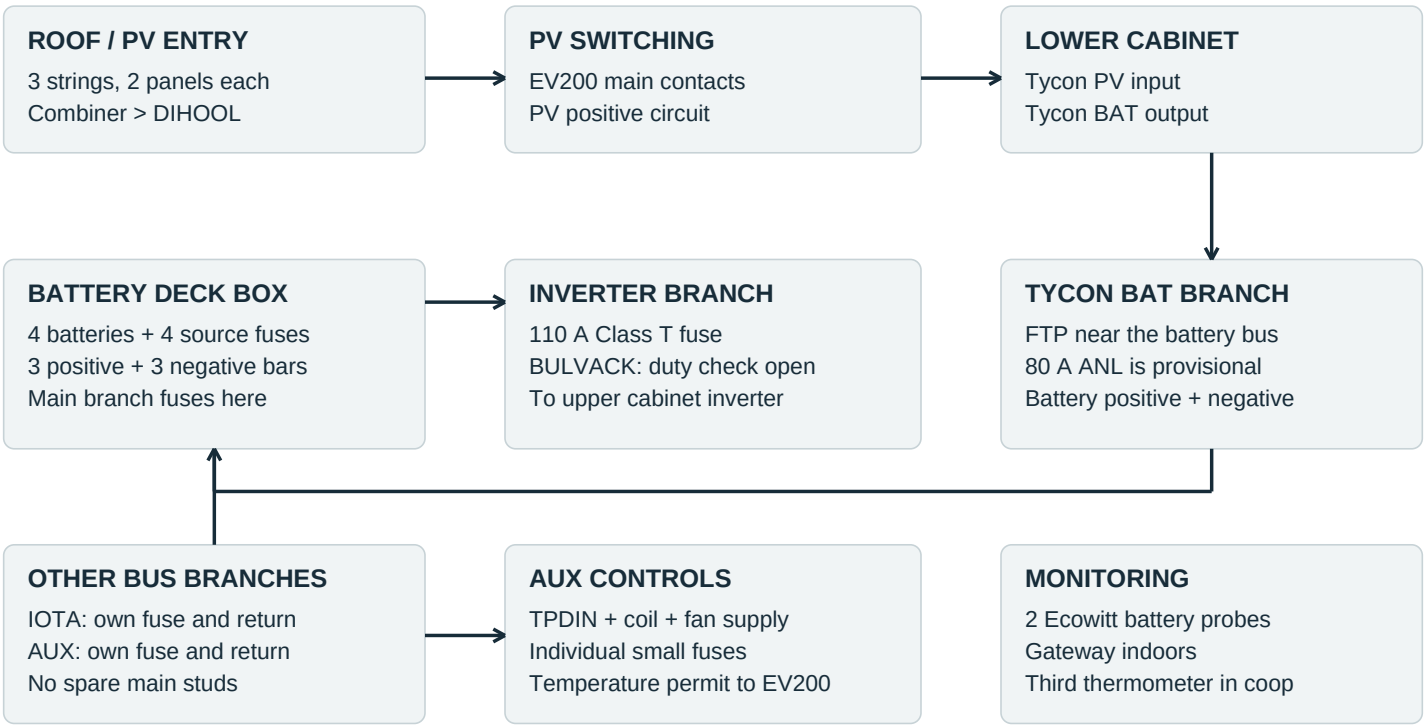
Electrical layout map

A detailed planning packet for the four-battery bank, solar charging, inverter, controls and new monitoring equipment.

Public revision 0.2 | 12 October 2026 | Proposed layout for review. This packet supports part placement and measurement. Final cable sizes, device coordination, cooling clearances and terminal instructions remain open. It is not a connection or energization release.



What changed in this map

The board is empty. You confirmed one spare positive bar and one spare negative bar. The proposed map therefore uses **three four-stud bars per polarity**. Each polarity has 12 studs, four jumper ends and eight branch connections. Each stud receives one cable lug.

Find it	Pages
Equipment and physical locations	2-3
Positive and negative stud maps	4-5
Fuses, cables and cabinet fit	6-7
Solar switching and temperature controls	8-10
AUX, IOTA, heaters, routing and grounding	11-12
Measurements, preparation, checks and sources	13-15

Equipment and status

Ownership, location and purpose are separate fields. An order does not establish receipt or installation.

Equipment / identifier	Status from the record	Planned location / purpose
Four LiTime 24 V 100 Ah packs, B1-B4	Owned; four-pack main bank retained	Deck box. Battery branches remain individually fused.
Six four-stud battery bus bars	Owned, including spare pair; board empty	Deck-box power board. P-A/B/C and N-A/B/C.
Seven ANL fuse holders	Seven recorded in older inventory; recount	Four batteries, Tycon, IOTA, AUX. Exact holder ratings pending.
150 A ANL battery fuses	User confirms battery fuses on hand	FB1-FB4. Verify four correct pieces and markings.
Blue Sea 5007100 + 5112	Ordered; block Oct 16-20, fuse Oct 19	FINV, 110 A Class T. Deck box near inverter bus takeoff.
BULVACK 80 A breaker, QINV	User confirms owned; installation unconfirmed	Inverter positive branch; switching/coordination check open.
TE EV200HAANA, 1-1618002-8	Ordered; delivery not recorded	KPV. Controlled interruption of PV positive.
Tycon TP-SC24-60N-MPPT	60 A model reported received and mounted; label still needed	Lower cabinet or revised mounting. Full array requires correct model.
TPDIN, DDR-30G-24, MC-9b, IOTA	Shown in Oct 4 lower-cabinet photo	Controls, fan supply, future AC switching, grid charger.
GW3000 + two WN34BL sensors	Ordered; expected Oct 12	Gateway indoors. Two battery-case monitoring probes.
Third thermometer	User confirms obtained for coop; model unspecified	Chicken coop. Separate from battery monitoring.
CAT6A couplers / AA kit	Couplers ordered. AA shipment unconfirmed	Network junctions / indoor charging. AA sensor compatibility unverified.

Other recorded orders include the 29.2 V bench charger, polycarbonate guard and standoffs, lever connectors, MC-9b AP/T-9 covers and 80 A ANL fuses. Recount delivered parts before buying duplicates. A third monitoring thermometer does not supply the battery-box air or outdoor monitoring position.

Battery-box placement

Arrange the board around the battery terminals and cable exit. Final coordinates depend on measured clearances and cable bends.

POWER BOARD: recorded 48 x 16 in; usable area needs measurement

Top view. FRONT is the service side. Blocks show zones, not scale.

N-A / N-B / N-C
Negative bars and return lugs

P-A / P-B / P-C
Positive bars and source fuses

CABLE EXIT / SERVICE
Class T and switch branch
FTP / FIOTA / FAUX
Choose side from actual route

FB1 / BATTERY 1
Short protected route to positive terminal

FB2 / BATTERY 2
Short protected route to positive terminal

FB3 / BATTERY 3
Short protected route to positive terminal

FB4 / BATTERY 4
Short protected route to positive terminal

SERVICE SPACE: wrench access, fuse removal, cable support and covers

Batteries are below the board. Their terminal positions control fuse placement.

Support board from structure. Keep access to battery terminals and sensor mounts.

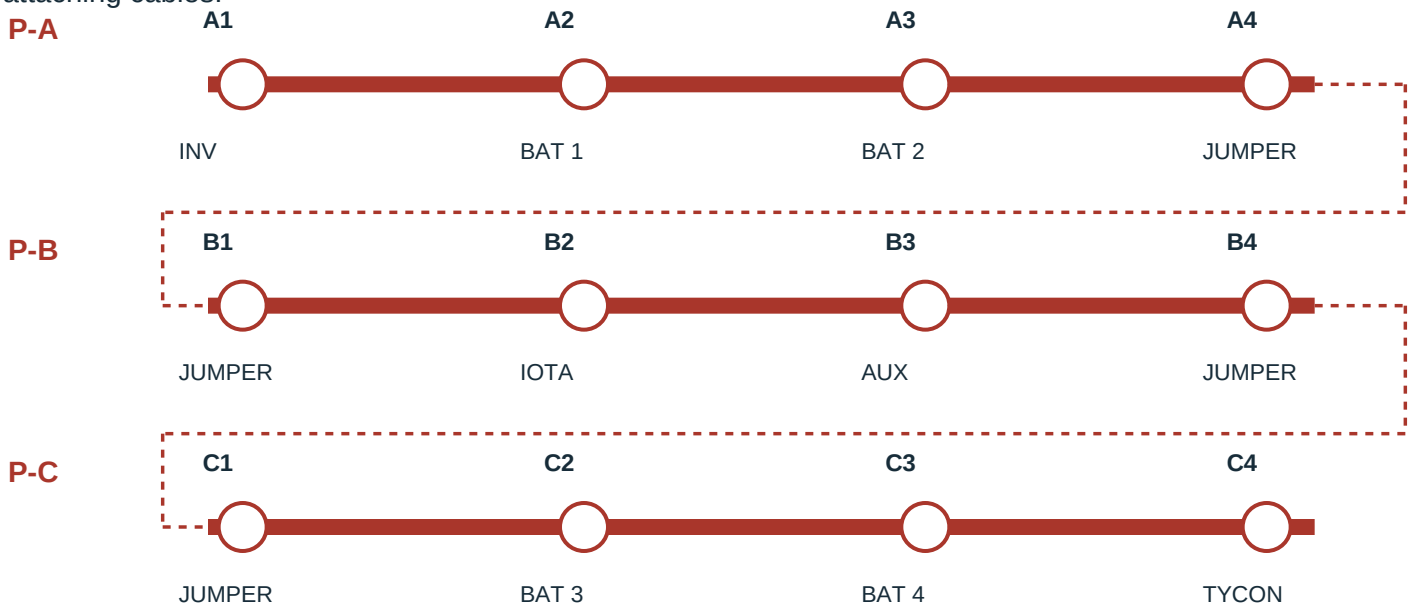
FRONT / lid-open service position

Layer / area	Layout requirement
Battery support	Four packs in the main bank. Recorded box interior: 53.3 x 26.5 x 24 in. Verify clear inside space after insulation and supports.
Power board	Recorded proposal: 48 x 16 in. Support it from the box structure. Do not assume battery lids can carry the board or cable loads.
Positive hardware	Keep P bars, FB1-FB4 and bus-side branch fuses covered. Short battery-to-fuse and bus-to-fuse routes control the unfused exposure.
Negative hardware	Place N bars in a separate clear zone. Prevent a dropped tool, loose cable end or guard fastener from bridging polarities.
Surface and guard	Current plan requires a noncombustible board face and a removable polycarbonate drop guard on insulating supports. Keep fuse covers accessible.
Sensors / future heat	Leave access to cold case faces and side-warmer positions. Keep sensor cables clear of high-current lugs, terminals and heater surfaces.

The recorded two-inch board standoff is a historical proposal. Measure the terminal, cable-lug and guard stack before retaining it. Insulation, heater attachment and lid movement can change the usable space.

Positive bus: every stud

Label the bars P-A, P-B and P-C. Number each bar 1 through 4 in a consistent physical direction before attaching cables.



Dashed paths: A4 to B1, and B4 to C1. One cable lug per stud.

Stud	Cable endpoint	Stud	Cable endpoint
A1	FINV to inverter positive	B3	FAUX to small-circuit fuse block
A2	FB1 from battery 1 positive	B4	J-PBC to P-C1
A3	FB2 from battery 2 positive	C1	J-PBC from P-B4
A4	J-PAB to P-B1	C2	FB3 from battery 3 positive
B1	J-PAB from P-A4	C3	FB4 from battery 4 positive
B2	FIOTA to IOTA positive	C4	FTP to Tycon BAT positive

Fuse position depends on the source

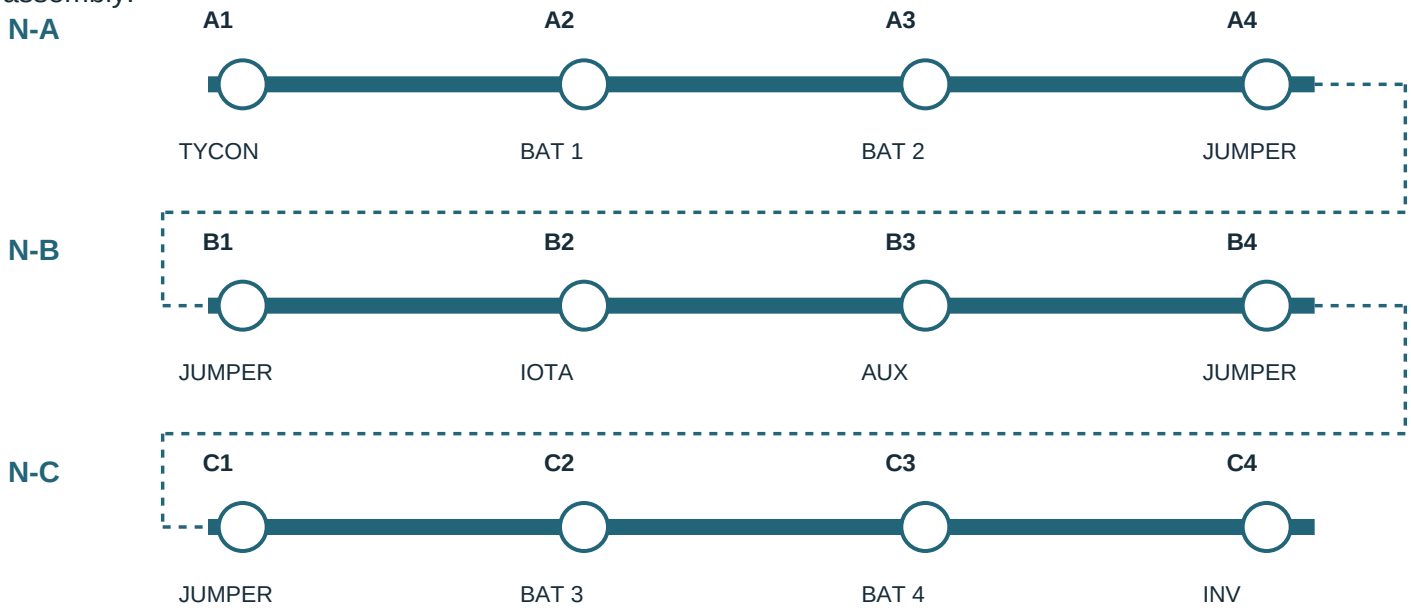
For a battery branch, the functional path is **battery positive > FB fuse > assigned P stud**. For the inverter or AUX branch, it is **P stud > branch fuse > equipment**. The Tycon and IOTA branch fuses sit near their P studs because the bank can supply fault current into a damaged charger cable.

Jumpers: J-PAB joins P-A4 to P-B1. J-PBC joins P-B4 to P-C1. Proposed cable is 2/0 copper, subject to the complete rating review. Use insulated flexible cable and supported lugs. No bare bolt or improvised metal strap is specified.

No free positive main-bus stud remains. A later high-current branch requires a revised distribution design. Additional small circuits can use suitable spare ways in the AUX block.

Negative bus: every stud

Label the bars N-A, N-B and N-C. These are circuit returns. The protective-earth bar is a separate assembly.



Dashed paths: A4 to B1, and B4 to C1. One cable lug per stud.

Stud	Cable endpoint	Stud	Cable endpoint
A1	Tycon BAT negative	B3	Small-circuit negative return block
A2	Battery 1 negative	B4	J-NBC to N-C1
A3	Battery 2 negative	C1	J-NBC from N-B4
A4	J-NAB to N-B1	C2	Battery 3 negative
B1	J-NAB from N-A4	C3	Battery 4 negative
B2	IOTA negative	C4	Inverter negative

Current sharing and isolation

The inverter uses P-A1 and N-C4. The Tycon uses P-C4 and N-A1. These opposite-end takeoffs reduce one source of unequal branch resistance. Make comparable battery leads equal in length, size and construction. The two center branches and linked-bar joints still affect current sharing; measure all four packs during charging and discharge.

Four jumpers total: two positive and two negative, requiring eight jumper-end lugs. Keep cable pairs together along their route where practical. Arrange bends so terminal studs carry no cable strain.

A jumper switch cannot isolate this whole bank. Batteries attach on both sides of each jumper. Opening a jumper splits the bank and can leave parts energized. A whole-bank disconnect needs a separate collection/distribution design. This map does not provide one.

Main fuses and cable schedule

The amp rating, fuse type, holder rating, conductor size and fault-interruption rating must all suit the branch.

ID / circuit	Positive path	Current proposal
FB1-FB4 / battery branches	B1-B4 positive > individual fuse > P-A2, P-A3, P-C2, P-C3	4 x 150 A ANL; 2/0 copper. Equal branch construction. Source-side fault protection requires review.
FINV / inverter	P-A1 > Blue Sea holder and fuse > QINV > inverter positive	110 A Class T; 2/0. QINV is the owned 80 A BULVACK, pending duty/coordination acceptance.
FTP / Tycon battery circuit	P-C4 > FTP > Tycon BAT positive; BAT negative > N-A1	80 A ANL and 4 AWG proposed. Manufacturer manual/spec-sheet conflict remains open.
FIOTA / grid-charger DC	P-B2 > FIOTA > IOTA positive; IOTA negative > N-B2	30 A ANL and 10 AWG proposed; verify output lead and terminal instructions.
FAUX / small circuits	P-B3 > FAUX > AUX positive block; AUX return > N-B3	30 A ANL and 10 AWG proposed. Each outgoing smaller circuit gets its own fuse.
J-PAB/PBC/NAB/NBC	Dedicated bar-to-bar studs, as pages 4-5	4 x 2/0 jumpers proposed. Ampacity, joints and bus ratings need review.

What to count on the bench

• Seven ANL holders: four for batteries, one Tycon, one IOTA and one AUX. The Class T uses its own holder.

• Four 150 A battery fuses, one proposed 80 A Tycon fuse, and **two** proposed 30 A fuses. An old 70 A Tycon fuse does not match revision B.

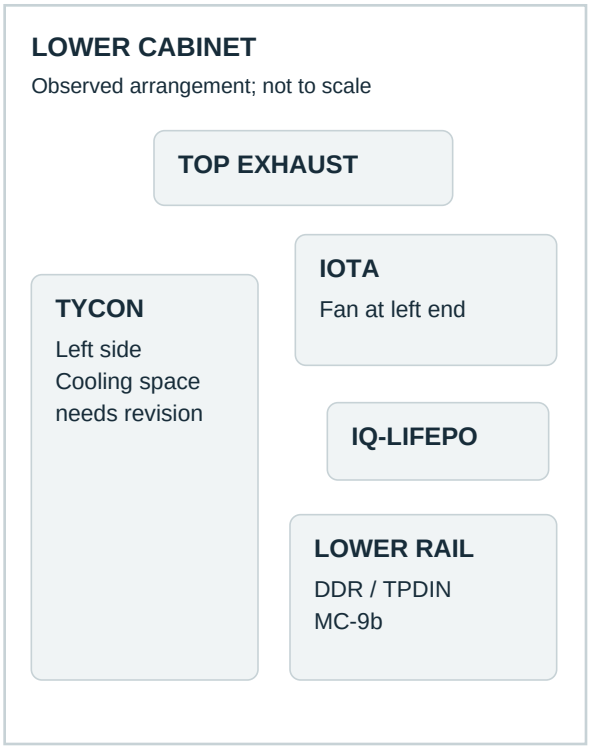
• Class T holder terminal studs are 1/4-20. Its cable lugs must match. Bus, battery, breaker and ANL studs need their own measured lug-hole sizes. [S2]

• Fuse coordination stays open. The Class T protects its inverter branch; it does not validate the other ANL fuses or a downstream breaker.

80 A is a design proposal, not a code conclusion. Tycon documents differ on the required battery fuse. Resolve this with the exact equipment instructions before selecting the final fuse and cable. The existing 80 A inverter device also does not support the inverter's full 3000 W rating.

Cabinet fit and new hardware

The October 4 photo establishes the arrangement. It does not establish that the required cooling and service clearances fit.



Observed in the lower cabinet

Tycon at left.
IOTA at upper right.
IQ-LIFEPO module below IOTA.
DDR, TPDIN and MC-9b on the lower rail.
Exhaust fan at top; side intake and right-side cable entries.

Unverified: mounted Tycon model, free depth, exact conductor routes and cooling envelopes.

Part / clearance	Measured or published basis	Layout consequence
Tycon TP60	245 mm body height + 200 mm above + 200 mm below = 645 mm, about 25.4 in. [S3/S4]	Recorded lower backplate is 19 in tall. Existing nominal cabinet height is also smaller than this envelope. Rework mounting/enclosure; a fan does not waive clearance.
EV200HAANA	About 3.17 x 2.45 in footprint; 2.85 in to stud tops before lugs and boots. [S1]	Reserve extra cable-bend, cover and door space. Chassis mount; no DIN clip is specified.
5007100 fuse holder	7.00 x 2.00 in footprint, 2.30 in covered height. [S2 drawing]	On battery board near inverter takeoff. Allow cover lift and two heavy-cable approaches.
IOTA / DDR	IOTA airflow space needs checking. DDR: 40 mm above, 20 mm below, 5 mm sides. [S7/S8]	Keep EV200 and cable bundles out of these spaces. Verify the IOTA fan-end clearance.

Preferred next layout check: reserve a suitable adjacent PV enclosure for KPV if it cannot fit. This option still requires its own fittings and protection. It does not resolve the Tycon's existing clearance conflict.

Solar power route

PV and battery wiring are different circuits. KPV interrupts the solar positive conductor upstream of the charge controller.

Segment	Connections and placement	Checks before connection
Roof strings S1-S3	Two panels in series per string. Three separate positive/negative pairs descend to MNPV3.	Retain 2S3P. Verify cold open-circuit voltage stays below the TP60's 100 V limit.
MNPV3 combiner	Each string positive through its own 15 A breaker to positive combining bus. Negatives to insulated PV-negative bar.	Three 15 A breakers user-confirmed. Exact breaker labels, torque and connector compatibility need records.
Manual disconnect QPV	Combined pair enters DIHOOL top/source pair. Bottom pair is controller side.	Keep conductor polarity consistent with delivered diagram. Verify actual lead markings.
Contactor KPV	DIHOOL output PV positive > EV200 main A1 (+) > main A2 (-) > Tycon PV positive.	Both main studs are in the POSITIVE path. The (-) marking identifies contact polarity, not a negative return.
PV negative	DIHOOL output PV negative > Tycon PV negative. It bypasses the EV200.	Do not connect this lead to the EV200 main A2 stud.
Tycon battery path	Tycon BAT positive > FTP > P-C4; BAT negative > N-A1.	BAT and PV terminals remain distinct. Full-array operation requires the verified 60 A controller.

Wire routes to reserve

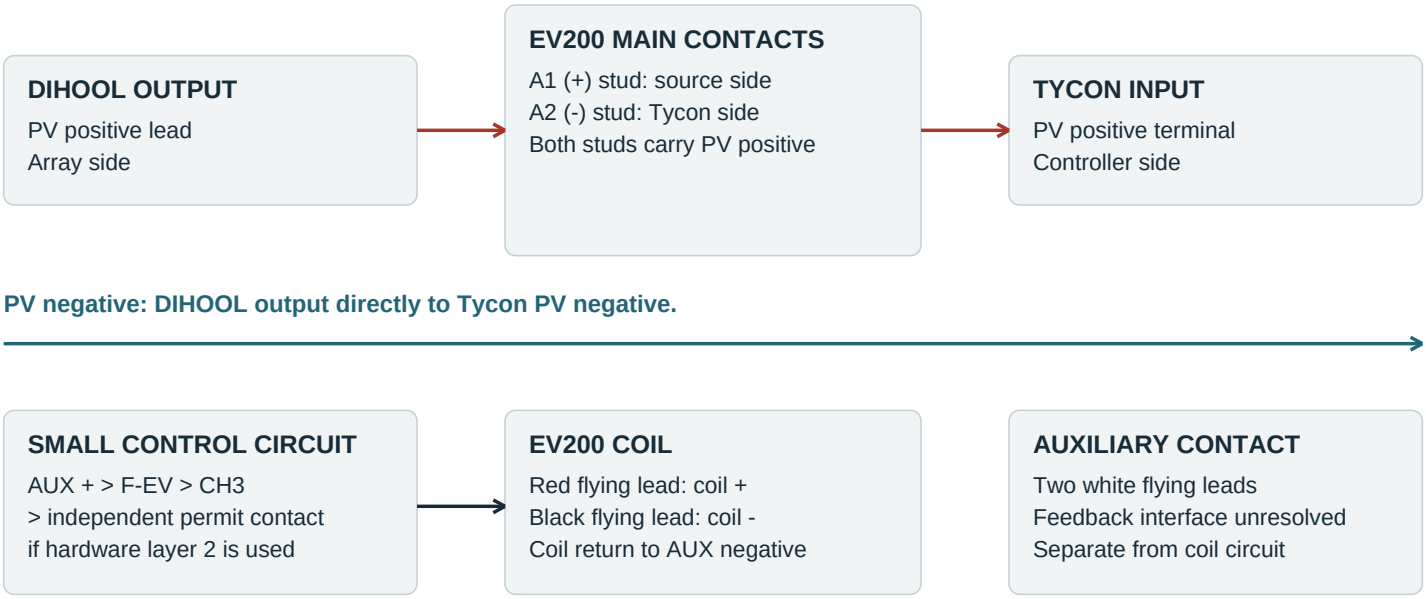
The current proposal is 10 AWG for each separate string and 8 AWG for the combined PV route. The two existing short combiner-to-DIHOOL jumpers are reported as 10 AWG. Their final acceptance depends on actual wire, terminal, grouping and temperature ratings. Keep them flagged in the cable schedule; do not treat their short length as an ampacity approval.

Route each circuit's positive and negative together. Give PV conductors a defined route separate from temperature and data wiring. Use rated entries and strain relief. The current conduit choice remains subject to measured cable diameter and fill.

Opening KPV leaves the array side live in sunlight. KPV is the automatic charging control; the manual service-isolation procedure must also account for PV, battery, grid and inverter sources.

EV200: power, coil and feedback

These are three separate functions. Identify every terminal from the delivered part before making a connection.



Connection	Proposed function	Detail to retain
Two large main studs	Carry switched solar positive	TE labels A1 (+) and A2 (-). Use correct current direction, covered studs and supported cables. [S1]
Red / black flying leads	Coil supply and return	Red positive, black negative. 9-36 V coil variant; built-in economizer. No extra coil suppression is specified by this map. [S1]
Two white flying leads	Normally open auxiliary contact	Feedback only. Minimum switching load is specified as 100 mA at 8 V. Direct high-impedance sensing needs an interface review. [S1]
Coil permit path	AUX positive > F-EV > TPDIN CH3 > independent hardware permit, if used > coil red	Return coil black to AUX negative. Final TPDIN terminal identification, contact duty and fault behavior remain open.

What the feedback can establish

The auxiliary contact can report contactor position under its specified conditions. The reviewed TE sheet does not establish a safety-rated mirror contact. Auxiliary state alone cannot prove zero charging current or guaranteed detection of welded main contacts. The commissioning procedure must verify actual charger behavior.

If layer 2 is the confirmed battery BMS, its protection occurs inside each battery. If layer 2 is a separate controller, it needs its own probe and permission contact. An external controller using KPV cannot open welded KPV main contacts. **TPDIN can also receive PoE power:** removing its DC feed alone may not remove all power.

Temperature control and probes

Protective sensing and monitoring use different equipment. The temperature rules below remain proposed until tested.

Sensor / device	Physical location	Role
T1 / TPDIN wired probe	Cold case location on the pack expected to be coldest; retained mechanically, insulated from terminals, away from warmers	Layer 1 charging permission. Attach to the battery case; ambient temperature alone is insufficient.
TM2 probe, if required	Separate probe on a different battery, chosen after the temperature survey	Independent hardware layer 2 if serial-specific LiTime BMS protection cannot be established.
Ecowitt E1 / E2	Two separate battery cases; place and label consistently for the temperature log	Monitoring and comparison. They do not switch charging.
Coop thermometer E3	Chicken coop, away from direct heater, sun and water exposure	Coop monitoring. Model and mounting details remain to be recorded.
Box-air and outdoor sensors	Two planned positions; sensors not established by current orders	Would help compare case temperature with enclosure and outside air.
Tycon temperature probe	Position and behavior per exact controller instructions	Separate from T1. Compensation behavior in lithium mode remains unresolved.

Layer 1 condition	Proposed result
T1 below 5 C / 41 F	Remove permission without the warm-up delay.
T1 above 8 C / about 46 F for 60 minutes	Permit restart only when layer 2 and all other required conditions are valid.
Between thresholds	State depends on previous state. Exact delay reset and restart behavior must be tested.
Probe fault, reboot or control power loss	Required result: charging off. Verify actual firmware and relay behavior on an isolated bench.

Cold-soak recovery remains open. A warm case cannot establish the coldest cell temperature. The one-hour delay is a proposal, not demonstrated core warm-up. One sensed pack also cannot prove that all four packs are warm. A detached probe may give a plausible reading; physical retention and multi-pack comparison matter.

Test immediate trip, continuous warm timing, timer reset, reboot and power restoration separately. A confirmed internal BMS can provide another interruption path; verify all four serial numbers. No battery is to be charged cold as a test.

AUX, IOTA and heater branches

The small fuse block distributes controls from one main-bus connection. IOTA charging and battery heating remain separate branches.

Small branch	Protection / wire proposal	Path and remaining check
F-TPD / TPDIN power	2 A; 18 AWG	AUX + > fuse > TPDIN power; return to AUX negative. Confirm exact power-input terminals. [S5]
F-SENSE / voltage estimate	1 A; 18-22 AWG	AUX + > F-SENSE > V1, with its specified reference to AUX negative. Reading includes AUX feeder voltage drop; compare with the battery bus.
F-EV / EV200 coil	5 A; 18 AWG	Through temperature permissions to coil. Check inrush, relay DC duty and maximum bank voltage.
F-DDR / fan supply	Final fuse unresolved; 16 AWG proposed	Rev B says 3 A. Mean Well recommends T6.3A/L250V for DDR-30G. Select suitable DC protection with wire and startup review. [S7]
Spare protected ways	Rating follows each selected load	Allow TM2 and future approved control circuits. Match block DC voltage, input current and terminal limits.

Place the AUX assembly near the controls

Proposed location: a covered small fuse block and negative return block near the TPDIN, in the lower cabinet or a suitable control enclosure. FAUX stays at P-B3 in the battery box. A protected positive feeder and negative return run between the boxes. F-SENSE uses the AUX block in this revision. Direct bus sensing would require a separately designed, protected takeoff; no main stud is spare.

IOTA grid charging, Phase 3

The IOTA DC path uses FIOTA and N-B2. Its AC path is an approved supply > MC-9b contactor > fixed receptacle > factory IOTA cord. Cold-charge permission must override the low-battery charging request. The MC-9b needs regulated 24 V; its verified coil range excludes the full charging-bus voltage. Final interface, AC terminals and supply allocation remain open. Keep grid and inverter AC circuits separate.

Battery heat, Phase 2

Proposed route: Tycon LOAD positive > feeder fuse > CH4 > individual heater fuses and high limits > warmers. Returns go to **Tycon LOAD negative**. Validate current, relay duty, voltage cutoff and extreme-cold discharge limits. One warm sensed pack must not stop required heat for a colder pack; all-pack demand or local regulation remains unresolved.

The DDR-30G-24 is reserved for the fan circuit. Its regulated output return remains separate from the battery return in the proposed isolated arrangement. Spare output capacity for other loads has not been proved.

Cable routes and grounding

Label both ends by circuit and endpoint. Route the actual cables before choosing lengths, glands or final lug angles.

Route ID	From / to	Reserve and identify
R1 / INV	Battery board to upper cabinet	2/0 positive and negative proposal. FINV and QINV at source-side branch. Right-side cabinet entries; bend and support space.
R2 / TP-BAT	Battery board to Tycon	4 AWG pair proposal. FTP near P-C4. Label BAT+ and BAT- at the controller.
R3 / IOTA-DC	Battery board to IOTA	10 AWG pair proposal. FIOTA near P-B2. Separate from the IOTA AC route.
R4 / AUX	Battery board to small-circuit block	10 AWG pair proposal. FAUX near P-B3. Negative feeder to N-B3.
R5 / SENSE + T1	Local AUX block / battery case to TPDIN	F-SENSE and reference return from AUX; reading includes feeder drop. Separate T1 cable from case. Keep clear of PV and AC terminals.
R6 / PV	DIHOOL to KPV to Tycon	8 AWG combined-route proposal; EV200 main stud polarity labeled. PV- bypasses KPV.
R7 / CONTROL	AUX / TPDIN to KPV; later MC-9b	Coil/control leads, strain relief and separated wiring path. Aux-contact interface still needs design.
R8 / DATA	House network to local controls	Ethernet and RS485 remain separate interfaces. CAT6A couplers provide cable joining; surge protection is a separate design.
R9 / PE	Approved house grounding system to EB1 and equipment	Permanent equipment bonds. Final conductor sizes and attachment points pending site verification.

Protective earth has its own map

Place EB1, the equipment-grounding bar, at the approved local bonding point. Connect the cabinets, doors and backplates, required equipment chassis terminals, combiner and array bonding system through the completed design. Protective earth carries no normal circuit return current. Do not switch or fuse PE. [L3]

No DC-negative-to-PE jumper or inverter neutral-to-PE jumper is assigned here. The exact equipment instructions and source arrangement must resolve those connections. Permanent solar equipment bonds must remain effective when the IOTA cord is unplugged.

The old ten-gland plate schedule predates these changes. Measure each gland's usable cable range and each cable's outside diameter. A fitting needs a suitable seal or multi-hole insert for its conductor arrangement. Reserve additional entries if the measured fill requires them.

Measurement and preparation sheet

Use the empty board for a dry layout. Keep all electrical sources disconnected while checking placement and making supports.

Record	Measurement or evidence needed	Result
Deck box and board	Clear inside dimensions; actual board size; support points; height above terminals; guard height; lid clearance	_____
All six bus bars	Model, DC rating, footprint, stud spacing, stud diameter/thread, approved torque and cover fit with jumpers	_____
Battery terminals	Position of each + and - post; case spacing; exact path to its fuse; terminal and lug access	_____
Fuse holders	ANL model/rating and seven-piece count; Class T body/cover clearance; lug and tool access	_____
Inverter branch	Full inverter model; QINV model/rating; cable path; both stud sizes; intended maximum office load	_____
Lower cabinet	Clear inside height/width/depth; door sweep; Tycon full label; airflow space; free side/backplate areas	_____
EV200 location	Delivered outline; hole centers; lug/boot envelope; cable bends; whether adjacent enclosure is required	_____
Control equipment	TPDIN hardware and firmware; probe model; exact relay labels; AUX block ownership/model; DDR markings	_____
Cable routes	R1-R9 lengths including bends and service access; actual outside diameters; gland count and seal ranges	_____
Manufacturer answers	LiTime BMS and heater approval; Tycon fuse/clearance/compensation; unresolved contact/coil ratings	_____

Work that can start now

• Recount and label the unconnected parts. Photograph markings before mounting hides them.

• Make paper outlines from the delivered parts. Lay out P and N zones, seven ANL holders, Class T holder, QINV and cable approaches on the board.

• Use flexible cord to trace each cable route. Leave room for cover removal, terminal access and the actual cable bend limits.

• Record two lengths for each battery branch: battery-to-fuse and fuse-to-bus. Compare the complete positive/negative branch construction before cutting.

• Take an overhead board photo and a straight-on cabinet photo with a ruler or tape. These determine the next dimensioned revision.

Checks before final assembly

The map makes the branches explicit. The checks below determine which proposed details can become installation instructions.

Check	Pass evidence
C1 / hardware ratings	Verified fuse and holder DC/fault ratings; bus current ratings; terminals, cable ampacity and protection coordinated. Recount fuse values.
C2 / inverter device	QINV load-switching and short-circuit coordination accepted, or a suitable replacement selected. Intended load within complete branch capability.
C3 / Tycon branch	Exact 60 A model identified; fuse-manual conflict resolved; cooling envelope fits; final battery and PV conductor schedule approved.
C4 / thermal protection	Layer 2 established. Coldest-pack assumption and recovery method validated. Thresholds, timer reset, startup and probe-fault states tested.
C5 / coil and feedback	CH3 DC duty/inrush check complete. EV200 coil wiring verified. Auxiliary load/interface designed; actual interruption check defined.
C6 / enclosure and route	All covers, cable supports, weather seals and service access fit. No cable or new device occupies required cooling space.
C7 / bonding and AC	Permanent PE design verified; inverter neutral arrangement resolved; IOTA supply and shared branch capacity accepted for Phase 3.
C8 / proof and records	Final terminal torque from each manufacturer recorded; configuration backup and labels complete; inspection and commissioning results retained.

Commissioning sequence to develop after C1-C7

Use an isolated, current-limited control test setup and simulated temperature inputs. Test probe open/short faults, reboot and loss of all control power. Verify timer cancellation when temperature falls during warming. Other rules and remote commands must not bypass temperature permission. Remove test connections afterward; never charge a cold battery as a test.

For power commissioning, use a written source-isolation and connection procedure for the final equipment. The Tycon requires battery connection before PV connection, and PV removal before its battery connection is removed. Inverter capacitor precharge requires its own verified method. Do not use the bus jumpers as routine switches.

Once approved and operating, measure each pack's charge and discharge current and inspect connection temperatures. Test solar and IOTA cessation separately and together in Phase 3. Measure each charger's output current; net bank current can hide continuing charge when loads run. Keep the results as a baseline.

Limits of this map: it does not include a whole-bank main disconnect, a final fault-current calculation, an approved AC terminal schedule, or a validated cold-soak warm-up time. These gaps are explicit so the drawing cannot be mistaken for a completed wiring release.

Sources and corrections

Local records establish the project state. Manufacturer documents establish component requirements. The new stud assignments are a proposal.

L1. Winter build plan revision B, 9 October; order and thermometer updates 11 October. WINTER-BUILD-PLAN-2026-10-09.md

L2. Project memory and electrical audit; current updates take precedence over historical layouts. CLAUDE.md; SOLAR-ELECTRICAL-AUDIT-2026-09-26.md

L3. PV entry and grounding design records. PV-COMBINER-PLAN-2026-09-27.md; SURGE-AND-BONDING-PLAN.md

S1. TE EV200HAANA product, linked EV200 data sheet and outline. Open source

S2. Blue Sea 5007100 product and dimensioned drawing. Open source

S3. Tycon TP30/TP60 manual. Open source

S4. Tycon TP30/TP60 specification sheet. Open source

S5. TPDIN WEB3 quick-install guide. Open source

S6. TPDIN WEB3 V2 release notes; match actual hardware before use. Open source

S7. Mean Well DDR DIN rail installation manual. Open source

S8. IOTA 24 V owner guide, manufacturer-authored document. Open source

Corrections carried into this packet

• Two linked bars provide six branch studs. The old map also named a third bar and stacked three charger/control leads. This proposal uses the owned third pair and eight separate connections.

• The old forced-air clearance exception is unsupported. The Tycon cooling envelope remains a physical fit conflict.

• A Class T fuse alone does not approve the BULVACK breaker or the other branch protection.

• Direct EV200 auxiliary-to-voltage-input wiring and welded-contact proof were overstated. The interface and test method remain open.

• The one-hour case-temperature delay remains unvalidated. Mean Well's fuse recommendation also needs reconciliation with revision B.

• Putting a switch in a bus jumper does not isolate a bank with batteries on both sides.

Technical review: 11 October 2026. Public edition: 12 October 2026. Public guide: view current layouts.